Forecast and alert sources

Every outside service El Nino Watch uses to forecast rain and river flow, to raise alerts, and to check those alerts against what has been observed in Somalia.

Each entry says what the app uses it for.

Checked against the app on 19 October 2026.

Official national source: Somali National Hydrological Services (MOEWR)

The Ministry of Energy and Water Resources publishes the national flood forecast the app shows as official alerts. ICPAC's regional forecasts, Google Flood Hub's river forecasts and GDACS event monitoring are not official warnings, and the app labels them that way.

Forecasts

6 sources

Open-Meteo Weather Forecast API

Open-Meteo

Daily and hourly rainfall forecasts for all 44 cities, and the 7-day rain forecast at eight points upstream in Ethiopia on the Shabelle and the Juba (Genale and Dawa), shown as river context on the Alerts tab. The app's server fetches them once and shares them with every visitor. These are computer model estimates, not station measurements.

Terms: Data under CC BY 4.0. The free API is for non-commercial use only, under 10,000 calls a day; commercial use needs a paid plan, which the app switches to when the owner sets its key.

ABOUT	`https://open-meteo.com/en/docs`
API	`https://api.open-meteo.com/v1/forecast`
PAID PLAN	`https://customer-api.open-meteo.com/v1/forecast`
TERMS	`https://open-meteo.com/en/terms`

Open-Meteo Flood API

GloFAS river discharge, via Open-Meteo

River discharge forecast for each city.

Terms: GloFAS data from the Copernicus Emergency Management Service: free, full and open access with the source credited, and no warranty. Open-Meteo's terms above also apply.

ABOUT	`https://open-meteo.com/en/docs/flood-api`
API	`https://flood-api.open-meteo.com/v1/flood`
PAID PLAN	`https://customer-flood-api.open-meteo.com/v1/flood`
TERMS	`https://ewds.climate.copernicus.eu/licences/cems-floods`

ICPAC Weekly Exceptional Rainfall Forecast

Also alerts

IGAD Climate Prediction and Applications Centre

Heavy, very heavy and extremely heavy rainfall forecasts for the week (above the 90th, 95th and 99th percentiles for that week of the year). Drives the Alerts tab and the rainfall layers on the flood map.

Terms: ICPAC does not state a reuse licence for these products. It takes data requests through its data request page.

ABOUT	`https://www.icpac.net/weekly-forecast/`
PORTAL	`https://eahazardswatch.icpac.net/`
DATA	`https://eahazardswatch.icpac.net/api/datasets/`
MAP LAYERS	`https://eahazardswatch.icpac.net/mapserver/`
TERMS	`https://www.icpac.net/data-request/`

ENSO Diagnostic Discussion

NOAA Climate Prediction Center

The current El Niño status and outlook.

Terms: No licence is stated on the CPC page. NOAA treats information on its servers as public domain unless marked otherwise, and asks to be credited.

ABOUT	`https://www.cpc.ncep.noaa.gov/products/analysis_monitoring/enso_advisory/ensodisc.shtml`
TERMS	`https://gml.noaa.gov/about/disclaimer.html`

Weekly Niño sea-surface temperature index

NOAA Climate Prediction Center

Weekly El Niño ocean temperatures.

Terms: No licence is stated on the CPC page. NOAA treats information on its servers as public domain unless marked otherwise, and asks to be credited.

DATA	`https://www.cpc.ncep.noaa.gov/data/indices/wksst9120.for`
TERMS	`https://gml.noaa.gov/about/disclaimer.html`

GEOGLOWS ECMWF Streamflow Service
GEOGLOWS, ECMWF

River flow forecast for the lower Juba near Jilib and Jamaame, shown as river context on the Alerts tab: the middle of the model's range for each of the next 15 days, and whether it is rising or falling. No flood thresholds are shown. The server checks it every 3 hours.

Terms: Data under CC BY 4.0, credited to the GEOGLOWS ECMWF Streamflow Service. ECMWF's terms also apply: outputs say they are based on ECMWF data, and ECMWF gives no warranty.

ABOUT	`https://geoglows.ecmwf.int/about`
API	`https://geoglows.ecmwf.int/api/v2/forecaststats/110483618?format=json`
TERMS	`https://geoglows.ecmwf.int/license`

The app also links to meteoblue's Galkayo forecast for comparison. It does not use meteoblue's data.

COMPARE `https://www.meteoblue.com/en/weather/today/gaalkacyo_somalia_59611`

Alerts and event monitoring

4 sources

Somali National Hydrological Services Official
Ministry of Energy and Water Resources (MOEWR)

The official national flood forecast: each region's daily flood level (advisory, watch or warning) for the active rainy season, with anticipatory-action pre-activation and activation. Shown as official alerts for the cities in each region.

Terms: Used with the Ministry's permission.

PORTAL `https://hydro.tekhub.so/`

Google Flood Hub
Google Research · Flood Forecasting API

River flood forecasts at river gauges, from Google's AI model. A gauge forecast at Flood Hub's Warning, Danger or Extreme level within 25 km of a monitored city becomes an alert for that city. A model forecast, not an official warning. Used once Google approves the app's access to the API; the data is under the CC BY 4.0 licence.

Terms: Data under CC BY 4.0.

PORTAL	`https://sites.research.google/floods/`
API	`https://floodforecasting.googleapis.com/v1/`
GUIDE	`https://developers.google.com/flood-forecasting`

GDACS

Global Disaster Alert and Coordination System

Flood and other disaster events affecting Somalia, shown as event monitoring.

Terms: Provided as is, without warranty; GDACS says its information is purely indicative and not for decision-making. No reuse licence is stated.

ABOUT	https://www.gdacs.org/
API	https://www.gdacs.org/gdacsapi/api/events/geteventlist/SEARCH
GUIDE	https://gdacs.org/Documents/2025/GDACS_API_quickstart_v1.pdf
TERMS	https://gdacs.org/About/termofuse.aspx

ICPAC Weekly Exceptional Rainfall Forecast

Listed under Forecasts

Rainfall alerts for each city, from the forecast above.

PORTAL	https://eahazardswatch.icpac.net/

Observations that support the alerts

5 sources

What has actually been measured or seen from satellites, used to check forecasts and alerts against conditions on the ground.

SWALIM river stations

FAO Somalia Water and Land Information Management

Observed river levels on the Juba and Shabelle, compared with flood thresholds.

Terms: SWALIM does not state a reuse licence: most records in its data catalogue list none. Ask FAO SWALIM before reusing the data elsewhere.

STATIONS	https://snrfa.faoswalim.org/stations/
FRRIMS	https://frrims.faoswalim.org/

Global Flood Monitoring (GFM)

Copernicus Emergency Management Service

Flood extent mapped from Sentinel-1 radar.

Terms: Copernicus Emergency Management Service terms: free and open access, commercial use included, with the service credited and no warranty.

ABOUT	https://extwiki.eodc.eu/GFM/PUM
MAP LAYERS	https://geoserver.gfm.eodc.eu/geoserver/gfm/wms
TERMS	https://extwiki.eodc.eu/GFM/Terms_and_conditions

Global Flood Product

NASA, served through NASA GIBS

3-day flood map layers from the MODIS and VIIRS satellites.

Terms: NASA Earth science data carry no restrictions on use (data from NASA-led missions are CC0); NASA asks to be credited.

ABOUT	`https://www.earthdata.nasa.gov/global-flood-product`
MAP LAYERS	`https://gibs.earthdata.nasa.gov/wmts/epsg3857/best/1.0.0/WMTSCapabilities.xml`
TERMS	`https://www.earthdata.nasa.gov/learn/use-data/data-use-policy`

Copernicus Data Space Ecosystem

Sentinel-1 and Sentinel-2 imagery

The latest satellite pass over each city, with a link to view the image.

Terms: Copernicus Sentinel data: free, full and open access under EU law, without warranty, with Copernicus credited.

BROWSER	`https://browser.dataspace.copernicus.eu/`
CATALOGUE	`https://stac.dataspace.copernicus.eu/v1/`
GUIDE	`https://documentation.dataspace.copernicus.eu/APIs/STAC.html`
TERMS	`https://sentinels.copernicus.eu/web/sentinel/-/free-access-to-copernicus-sentinel-satellite-data`

SWALIM flood maps

FAO SWALIM spatial data catalogue

Reference layers: terrain flood susceptibility (30 m), past flood areas 2013–2023, Shabelle riverbank breakages (August 2026 assessment) and Juba riverbank breakages (March 2026 assessment).

Terms: SWALIM does not state a reuse licence: most records in its data catalogue list none. Ask FAO SWALIM before reusing the data elsewhere.

SUSCEPTIBILITY	`https://spatial.faoswalim.org/catalogue/#/dataset/609`
PAST FLOODS	`https://spatial.faoswalim.org/catalogue/#/dataset/295`
SHABELLE	`https://spatial.faoswalim.org/catalogue/#/dataset/607`
JUBA	`https://spatial.faoswalim.org/catalogue/#/dataset/481`

Audio bulletin

Somali and English

A short spoken bulletin at 06:30, 12:30 and 18:30 East Africa Time, in Somali and English, first on the home page. It is written by a fixed template from the sources on this page and read by a computer voice (Google Gemini). It is not an official warning: official flood warnings come only from MOEWR.

El Nino Watch audio bulletin

Audio

El Nino Watch · read by Google Gemini (voice Puck)

Built from MOEWR's official flood levels, SWALIM's Juba and Shabelle river gauges, Open-Meteo's rain forecast for the 44 cities, ICPAC's weekly heavy-rain outlook, NOAA's El Niño status and news items in the bulletin's own language, with the source named. Nothing is machine-translated. The last seven days of bulletins are kept, and each can be saved as MP3.

LISTEN	`elninowatch.so/#podcast`
RSS, SOMALI	`https://elninowatch.so/podcast.xml`
RSS, ENGLISH	`https://elninowatch.so/podcast-en.xml`

Public health guidance

10 sources

The public health alert and the films on the Diseases tab are based on these World Health Organization documents. They are guidance, not outbreak reports.

Diseases from dirty water Film

El Nino Watch film · English, with Somali captions

A narrated overview of cholera, dysentery, typhoid and hepatitis A and E: how they spread, how to prevent them and when to get help. It is based on the WHO fact sheets in this list.

FILM `elninowatch.so/#waterborne-film`

Cholera: protect your family Film

El Nino Watch film · English and Somali

A narrated film on preventing cholera: safe water, handwashing, latrines, safe food and ORS. It plays in the reader's language and is based on the WHO cholera fact sheet and WHO's floods guidance below.

FILM `elninowatch.so/#health-film`

Malaria: protect your family Film

El Nino Watch film · English; Somali coming

A narrated film on preventing malaria: treated bed nets, removing standing water, stopping bites, and testing a fever the same day. It is based on the WHO malaria fact sheet and WHO's floods guidance below.

FILM `elninowatch.so/#malaria-film`

Dysentery and typhoid: protect your family Film

El Nino Watch film · English; Somali coming

A narrated film on dysentery and typhoid: the signs of each, safe water, handwashing, safe food, latrines, ORS, and medicine only from a health worker. It is based on the WHO typhoid and diarrhoeal disease fact sheets and WHO's floods guidance below.

FILM `elninowatch.so/#dysentery-film`

Hepatitis A and E: protect your family Film

El Nino Watch film · English; Somali coming

A narrated film on hepatitis A and E: the signs, why hepatitis E is dangerous in pregnancy, safe water, handwashing, latrines, safe food, and seeing a health worker. It is based on the WHO hepatitis A and E fact sheets and WHO's floods guidance below.

FILM `elninowatch.so/#hepatitis-film`

Global Public Health Situation Analysis, El Niño 2026

World Health Organization (WHO)

Lists cholera and other waterborne disease, dengue, malaria and Rift Valley fever as high or very high risks for Somalia (printed page 14).

REPORT `https://cdn.who.int/media/docs/default-source/meeting-reports/global-public-health-situation-analysis-el-nino-2026_.pdf`

Somalia Health Emergency Appeal 2025
World Health Organization (WHO)

Describes cholera in Somalia, and malaria and dengue spreading to new districts as a direct result of flooding.

APPEAL https://www.who.int/publications/m/item/somalia--who-health-emergency-appeal-2025

Floods: how to protect your health
World Health Organization (WHO)

Explains contaminated drinking water, mosquitoes after floods, and preventing cholera, typhoid and malaria.

Q&A https://www.who.int/news-room/q-a-detail/how-do-i-protect-my-health-in-a-flood

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Water, sanitation and health in humanitarian emergencies
World Health Organization (WHO)

Covers infections linked to poor sanitation: diarrhoeal diseases, dysentery, hepatitis A and E, and skin infections.

PAGE https://www.who.int/teams/environment-climate-change-and-health/water-sanitation-and-health/environmental-health-in-emergencies/humanitarian-emergencies

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Rift Valley fever
World Health Organization (WHO)

Links outbreaks in East Africa to El Niño heavy rainfall, and explains infection from livestock and mosquitoes and how to prevent it.

FACT SHEET https://www.who.int/news-room/fact-sheets/detail/rift-valley-fever

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Cholera
World Health Organization (WHO)

Explains contaminated food and water, rapid dehydration, ORS treatment, sanitation and vaccination.

FACT SHEET https://www.who.int/news-room/fact-sheets/detail/cholera

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Diarrhoeal disease
World Health Organization (WHO)

Explains how diarrhoea spreads through contaminated food and water, that bloody diarrhoea is dysentery, and treatment with ORS and zinc, with continued feeding and breastfeeding.

FACT SHEET https://www.who.int/news-room/fact-sheets/detail/diarrhoeal-disease

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Typhoid
World Health Organization (WHO)

Explains that typhoid spreads through contaminated food and water, causes a prolonged high fever, is treated with antibiotics and can be prevented by vaccination.

FACT SHEET https://www.who.int/news-room/fact-sheets/detail/typhoid

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Hepatitis A
World Health Organization (WHO)

Explains how hepatitis A spreads through contaminated food and water, its signs, including yellow eyes and skin and dark urine, and the vaccine.

FACT SHEET https://www.who.int/news-room/fact-sheets/detail/hepatitis-a

Hepatitis E
World Health Organization (WHO)

Explains that hepatitis E spreads mainly through contaminated drinking water and is especially dangerous for pregnant women.

FACT SHEET `https://www.who.int/news-room/fact-sheets/detail/hepatitis-e`

Privacy

What El Nino Watch keeps about you, and who else sees what. There are no accounts, no cookies and no advertising, and nothing is sold.

Visit counts

Each page load is counted, with the tab, city, language, type of device and the country the connection comes from, and so is each play of the audio bulletin. No cookies, internet addresses or browser details are stored, and browsers that ask not to be tracked are not counted. The totals are shown on the visitor statistics page.

Community reports

A reviewer checks each new report before it appears. Once approved, reports and comments are public: what you write, the name or alias you give (or Anonymous), any location you add, and your photos or videos. Location and device details stored inside photos and MP4 videos are removed on upload. Your internet address is held in memory for up to 10 minutes to limit abuse, and is never saved. The browser that sent a report can delete it, using a receipt kept in that browser, and a reviewer can remove it.

WhatsApp alerts

If you sign up, your WhatsApp number, the cities you follow, your language and whether you asked for the audio bulletin are kept only to send those messages. Messages go through Meta's WhatsApp service. Replying STOP deletes them.

Saved in your browser

Your language, home city, added cities, report receipts, alerts already shown, city manager review stages and private notes are saved in your own browser, not sent to El Nino Watch. Clearing the site's data in your browser removes them.

Other services

Forecasts, alerts and news are fetched by the El Nino Watch server, so those services do not see you. The audio bulletins are written from those sources and voiced by Google's Gemini service; nothing about you is sent to it. The flood map's background (OpenStreetMap or Esri), its layers and the place-name search for added cities load directly from their publishers, who see your internet address as any website does. Share buttons contact X or Facebook only when you tap them. The site is hosted on bunny.net, which handles every request as any web host does.

Questions about privacy: contact@mandekconsulting.com

Not listed: the base map and elevation services (Natural Earth, OpenStreetMap, Esri World Imagery, Open-Meteo Elevation and place-name search), which draw the map but do not forecast or alert.

Map backgrounds: Natural Earth is in the public domain. OpenStreetMap data are © OpenStreetMap contributors under the Open Database License, and its map tiles are used

under the OpenStreetMap Foundation’s tile usage policy. Esri World Imagery is used under Esri’s terms.

Community reports in the app come from residents and are unverified.